

Tubular Flow Reactor Project Summary Report

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The goal of the experiment was to determine the effect of flow rate on the conversion of the reaction between sodium hydroxide and ethyl acetate. For this experiment, the flow rates 0.25 GPM, 0.45 GPM, and 1 GPM were tested to observe the effect different flow regimes have on conversion. As part of this experiment, residence time distribution models (RTDs) were used to compare the results obtained from experimentation.

The tubular flow reactor consists of, two reagent storage tanks, two equal sized reagent holding tanks, one neutralization tank, two conductivity meters placed on the entrance and exit of the tubular reactor, two pumps for mixing the reagents, one pump to push fluid through the reactor, three control valves and flow meters to monitor flow of reagents into the reactor. To perform the experiment, solutions of 0.08 M and 0.1 M of sodium hydroxide and ethyl acetate were made, respectively. A sample of pure sodium hydroxide is drawn off to be titrated and its initial concentration calculated. Both reagents are fed into the reactor at equal flow rates. Each flow rate was reacted at 175% of its residence time to ensure steady state. After each flow rate was tested, the holding tanks were closed off and the reagents in the reactor were left to react until completion, after which the system was flushed.

By using conductivity data and the initial concentration of sodium hydroxide, the concentration of sodium hydroxide can be calculated and plugged into the equation below to determine the concentration.

$$X = 1 - \frac{C_{\text{NaOH}}}{C_{\text{NaOH},0}}$$

To ensure that the reported conversion only accounts for conversion of the flow rate in question, the average is taken after one residence time, the average time a particle spends in a

reactor. Due to time constraints and cost of materials, only one trial could be ran, so the error reported on the experimental conversions is due to systematic error.

For the theoretical conversion, the flow regime dictated the model used. For turbulent flow, an ideal plug flow reactor and RTD model were used. For laminar flow, a laminar flow reactor and RTD model were used. Due to the unpredictability of transition flow, no model exist specifically for transition. So, using the laminar flow and plug flow reactor models, a range of conversions was determined. Because RTD models have bound of integration between 0 and infinity, python was used to determine the theoretical conversion [1]. A complete summary of data collected is presented in Table 1.

Table 1: Summary of experimentally obtained conversion compared to the conversion calculated by the appropriate Residence Time Distribution Model

Flow Regime (Flow rate, gpm)	Conversion		Theoretical Model
	Experimental	Theoretical	
Laminar (0.25)	0.491 ± 0.00669	0.449	Laminar ¹
Transition (0.45)	0.382 ± 0.00661	0.311	Laminar ²
		0.347	Turbulent
Turbulent (1.00)	0.221 ± 0.00823	0.188	Turbulent

To answer the original objective posed at the start of this experiment, as flow rate through a tubular reactor increases, the conversion of a reaction decreases. This trend matches the trend of the theoretical models used. Laminar flow will yield higher conversion than that of turbulent flow. So, depending on the application a tubular reactor is used in, if a high conversion is needed, small flow rates must be used. If small flow rates are not achievable, a recycle stream may need to be used and unused reagents fed back into the reactor.

¹Estimated error of $4.841 \cdot 10^{-10}$.

²Estimated error of $5.937 \cdot 10^{-11}$.

Appendix

Sample Calculations

Experimental

Calculate the internal liquid volume of the cylindrical reactor. Use the measured length l and radius r and substitute numeric values to give the reactor volume in cubic inches, which is then converted to gallons for comparison with flow rates. This volume is needed to compute residence time and to relate liquid volume to flow.

$$V_r = \pi \cdot l \cdot r^2$$

$$V_r = \pi \cdot (952 \text{ in}) \cdot (0.25 \text{ in})^2$$

$$V_r = 186.9 \text{ in}^3 = 0.809 \text{ gal}$$

Compute the mean residence time (average time a fluid element spends in the reactor) by dividing the reactor volume V_r by the volumetric flow rate $\dot{V}$ (gallons per minute) that is being tested for that trial. The residence time is used to determine the appropriate reaction times and to compare them with experimental sampling times, so all relevant data is collected for the trial.

$$\tau = \frac{V_r}{\dot{V}}$$

$$\tau = \frac{0.809 \text{ gal}}{0.45 \text{ GPM}}$$

Multiply the residence time by the factor 1.75 to get the target sampling/processing time used in the experiment. For this case, to get the time a reaction is considered effectively complete. This gives a time in seconds consistent with the residence time units.

$$\tau = 1.80 \text{ min} = 107.9 \text{ sec}$$

$$1.75 \cdot \tau = 188.8 \text{ sec}$$

The conductivity reading from the solution is used as the calibration signal K . Species A is the limiting reactant, NaOH, while species B is ethyl acetate. To convert the raw conductivity measurement into the actual concentration of species A, the signal is normalized between the baseline conductivity K_∞ representing the fully reacted state, and the reference conductivity

K_0 , representing the initial, unreacted state. This linear calibration relationship allows conductivity values to be directly converted into concentrations.

$$C_A = \frac{K - K_\infty}{K_0 - K_\infty} \cdot C_{A,0}$$

$$C_A = \frac{6915.138 \frac{\mu S}{cm} - 6642.424 \frac{\mu S}{cm}}{7455.158 \frac{\mu S}{cm} - 6642.424 \frac{\mu S}{cm}} \cdot 0.0068167 \frac{mol}{L}$$

Calculate the average fractional conversion $\bar{X}$ of reactant A time t using the concentration C_A that is found above. This gives the fraction of A that has reacted, where $\bar{X} = 0$ means no conversion and $\bar{X}$ means complete conversion. This formula was used to calculate conversion for each second the trial was running and averaged the data points for each flow regime together. Below is an example of one of those points for the transition flow regime.

$$\bar{X} = 1 - \frac{C_A}{C_{A,0}}$$

$$\bar{X} = 1 - \frac{0.02287 \frac{mol}{L}}{0.068167 \frac{mol}{L}}$$

$$\bar{X} = 0.664$$

The initial concentration of species A was calculated by making an aqueous solution using the primary standard potassium hydrogen phthalate (KHP). The amount of volume displaced in the burette was multiplied by the molarity of the KHP solution, specifying the moles in the reactant A sample. The number of moles can be divided by the volume of sample, resulting in the initial concentration of reactant A.

Theoretical

The rate constant is calculated using the Arrhenius equation, which relates the frequency factor, activation energy, universal gas constant, and reaction temperature. This determines the specific rate constant for the reaction undergone in this unit. Both K_0 and E_A were sourced at room temperature [2].

$$k = k_0 \cdot e^{-\frac{E_A}{R \cdot T}}$$

$$k = 1.05 \cdot 10^3 \frac{\text{m}^3}{\text{mol} \cdot \text{s}} e^{\frac{-39.8 \frac{\text{kJ}}{\text{mol}} \cdot 1000 \frac{\text{J}}{\text{kJ}}}{8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}} \cdot 295.49 \text{K}}}$$

$$k = 9.283 \cdot 10^{-5} \frac{\text{m}^3}{\text{mol} \cdot \text{K}}$$

The overall conversion is expressed as an integral of the instantaneous conversion weighted by the residence time distribution (RTD). This accounts for the distribution of residence times in the reactor and provides the expected overall conversion. The RTD is dependent on the flow regime the molecules are in.

$$\bar{X} = \int_0^\infty X(t) E(t) dt$$

$$X(t) = \frac{C_{B,0} \cdot (1 - e^{-k \cdot (C_{B,0} - C_{A,0}) \cdot t})}{C_{B,0} - C_{A,0} \cdot e^{-k \cdot (C_{B,0} - C_{A,0}) \cdot t}}$$

Laminar Flow

For laminar flow, the residence time distribution is uniform between half the mean residence time and infinity. Substitute this distribution into the general conversion integral gives the theoretical conversion under laminar flow conditions.

$$E(t) = \begin{cases} 0, & t < 0 \\ \frac{\tau^2}{2t^3}, & t \geq \frac{\tau}{2} \end{cases}$$

$$\bar{X} = \int_{\frac{\tau}{2}}^\infty \frac{\tau^2}{2t^3} \cdot \frac{C_{B,0} \cdot (1 - e^{-k \cdot (C_{B,0} - C_{A,0}) \cdot t})}{C_{B,0} - C_{A,0} \cdot e^{-k \cdot (C_{B,0} - C_{A,0}) \cdot t}} dt$$

$$\bar{X} = \int_{\frac{\tau}{2}}^\infty \frac{\tau^2}{2t^3} \cdot \frac{\frac{0.1}{2} \text{M} \cdot 1 - e^{-9.283 \cdot 10^{-5} \frac{\text{m}^3}{\text{mol} \cdot \text{s}} \cdot \frac{1000 \text{L}}{1 \text{m}^3} (\frac{0.1}{2} \text{M} - \frac{0.076854}{2} \text{M}) \cdot t}}{\frac{0.1}{2} \text{M} - \frac{0.076854}{2} \text{M} \cdot e^{-9.283 \cdot 10^{-5} \frac{\text{m}^3}{\text{mol} \cdot \text{s}} \cdot \frac{1000 \text{L}}{1 \text{m}^3} (\frac{0.1}{2} \text{M} - \frac{0.076854}{2} \text{M}) \cdot t}} dt$$

$$\bar{X} = 0.491$$

Turbulent Flow

For turbulent flow, all fluid elements are assumed to have the same residence time equal to the mean residence time. Therefore, the overall conversion reduces to evaluating the

instantaneous conversion at that time only. Additionally, transitional flow is also modeled using the equation below.

$$\bar{X} = \int_0^\infty X(t)E(t) dt = \begin{cases} 0 & \text{if } t \neq \tau \\ X(\tau) & \text{if } t = \tau \end{cases}$$

$$\bar{X} = \frac{\frac{0.1}{2}\text{M} \cdot \left(1 - e^{-9.283 \cdot 10^{-5} \frac{\text{m}^3}{\text{mol} \cdot \text{s}} \cdot \frac{1000\text{L}}{1\text{m}^3} \left(\frac{0.1}{2}\text{M} - \frac{0.076854}{2}\text{M}\right) \cdot 48.55 \text{ sec}}\right)}{\frac{0.1}{2}\text{M} - \frac{0.076854}{2}\text{M} \cdot e^{-9.283 \cdot 10^{-5} \frac{\text{m}^3}{\text{mol} \cdot \text{s}} \cdot \frac{1000\text{L}}{1\text{m}^3} \left(\frac{0.1}{2}\text{M} - \frac{0.076854}{2}\text{M}\right) \cdot 48.55 \text{ sec}}}$$

$$\bar{X} = 0.221$$

Systematic Error

All conductivity readings were measured using the Model 54eC conductivity HART analyzer. The sensor used in the analyzer is a contacting conductivity sensor. For contacting conductivity sensors, the analyzer is accurate within $\pm 0.5\%$ of the reading as reflected in the calculations below [3], [4]:

$$\Delta_{X,\text{Tran}} = \sqrt{\left(\frac{\partial X}{\partial k} \cdot \delta_k\right)^2 + \left(\frac{\partial X}{\partial k_\infty} \cdot \delta_{k_\infty}\right)^2 + \left(\frac{\partial X}{\partial k_0} \cdot \delta_{k_0}\right)^2}$$

$$\Delta_{X,\text{Tran}} = \sqrt{\left(-\frac{1}{k_0 - k_\infty} \cdot \delta_k\right)^2 + \left(\frac{1}{k_0 - k_\infty} - \frac{k - k_\infty}{(k_0 - k_\infty)^2} \cdot \delta_{k_\infty}\right)^2 + \left(\frac{k - k_\infty}{(k_0 - k_\infty)^2} \cdot \delta_{k_0}\right)^2}$$

$$\Delta_{X,\text{Tran}} = \sqrt{\left(-\frac{1}{7912.609 \frac{\mu\text{S}}{\text{cm}} - 2809.877 \frac{\mu\text{S}}{\text{cm}} \frac{\mu\text{S}}{\text{cm}}} \cdot \delta_k\right)^2 + \left(-\frac{1}{7912.609 \frac{\mu\text{S}}{\text{cm}} - 2809.877 \frac{\mu\text{S}}{\text{cm}}} - \frac{5318.821 \frac{\mu\text{S}}{\text{cm}} - 2809.877 \frac{\mu\text{S}}{\text{cm}}}{\left(7912.609 \frac{\mu\text{S}}{\text{cm}} - 2809.877 \frac{\mu\text{S}}{\text{cm}}\right)^2} \cdot \delta_{k_\infty}\right)^2 + \left(\frac{5318.821 \frac{\mu\text{S}}{\text{cm}} - 2809.877 \frac{\mu\text{S}}{\text{cm}}}{\left(7912.609 \frac{\mu\text{S}}{\text{cm}} - 2809.877 \frac{\mu\text{S}}{\text{cm}}\right)^2} \cdot \delta_{k_\infty}\right)^2}$$

$$\Delta_{X, \text{Tran}} = \sqrt{\left(-0.000195973 \cdot \left(0.005 * 5318.821 \frac{\mu S}{\text{cm}}\right)\right)^2 + \left(0.000099615 \cdot \left(0.005 * 2809.877 \frac{\mu S}{\text{cm}}\right)\right)^2 + \left(0.000096357 \cdot \left(0.005 * 7912.609 \frac{\mu S}{\text{cm}}\right)\right)^2}$$

$$\Delta_{X, \text{Tran}} = 0.006607$$

References

- [1] J. Schlabach, *CHE412-Tubular-Flow-Reactor-Analysis*. (2025). Github. [Online]. Available:
<https://github.com/JoeGamer2120/CHE412-Tubular-Flow-Reactor-Analysis/blob/main/ModelInt.py>
- [2] A. Mendes, L. Madeira, F. Magãlhaes, and J. Sousa, “An Integrated Chemical Reaction Engineering Lab Experiment,” *Chemical Engineering Education*, 2006.
- [3] Emerson Process Management, “ENDURANCE® General Purpose Conductivity Sensors,” 2006.
- [4] Emerson Process Management, “Conductivity/Resistivity HART® Analyzer/Controller,” 2006.